

DESIGN SYSTEM

Design system reference for the portal

How to build screens that match the live reference at nikkortrinidad-cpu.github.io/flizow. Philosophy first, then tokens, then component patterns, then the per-screen audit checklist. Pair this with the auto-generated icon manifest ([docs/feature-specs/icon-manifest.md](#)) for the per-page Heroicon lookup.

This document is interactive. Tick each item in the audit checklist as you self-review a screen, then save the PDF. Run the audit on every page, modal, and feature before you ship it.

Overview

How to build screens that match the live reference. Walks through the design philosophy first (the belief, the 10 principles, the 10-question audit you run on every screen before shipping), then the concrete token reference (spacing, type, color, motion), then the component patterns we use (buttons, modals, dropdowns, chips, cards), then where to find the per-page Heroicon manifest. Read it once front-to-back and you'll have every answer you'd otherwise have to ask for. The live build at nikkortrinidad-cpu.github.io/flizow is the visual source of truth; this doc explains the rules behind what you see there.

Core belief

Every design decision filters through one sentence:

"The user came here to get something done. Get out of their way and help them finish."

Before designing any screen, ask: does this match the belief? If not, cut it. The principles below are derivations of this one idea — every rule traces back here.

The 10 principles

Working defaults. Each one has a reason; break them only when the reason no longer holds (see "When to break the rules" at the end of this section).

1

Clarity beats cleverness.

Every element has one obvious purpose. Plain words: "Save changes" not "Commit." Contrast meets WCAG AA. Never use color alone to signal meaning.

2

The interface defers to the content.

Content gets the biggest type, strongest contrast, most space. Chrome (nav, toolbars, panels) uses muted colors and smaller weights. If chrome fights content for attention, redesign the chrome.

3

Direct manipulation wherever possible.

Edit in place. Click the thing you want to change, not a button labeled "Edit." No pencil icons; cursor change + hover tint + ring carry the affordance.

4

Every action gets feedback.

Click feedback within 100ms. Loading states for any action over 200ms. Confirmation messages for saved changes. Motion matches physics: ease-out, not linear.

5

Forgiveness is non-negotiable.

Cmd+Z works. Soft delete before hard delete with a 90-day grace period. Confirmation dialogs only for truly destructive unrecoverable actions. Autosave drafts.

- 6 Consistency is the contract.**
One button style per hierarchy level. Same word for the same concept. Same icon for the same action. Fixed spacing and type scale used everywhere.
- 7 Hierarchy through typography, not decoration.**
Build a type scale and stay on it. Use weight and size for hierarchy before reaching for color or borders. One typeface (Inter). Body line length 45-70 characters.
- 8 Motion explains, it doesn't decorate.**
Every animation has a reason. Duration 150-400ms. Spring or ease-out, not linear. Respect prefers-reduced-motion: kill the animation or make it instant.
- 9 Accessibility is a layer, not a feature.**
Semantic HTML first (button, nav, main). Keyboard nav works for every interactive element. Visible labels on every form field. Alt text on every image. Honor system settings: dark mode, reduced motion, font size.
- 10 Speed is a design decision.**
First paint under 1.5s on mid-range phone. No layout shifts after initial load (CLS under 0.1). Optimistic UI for actions that usually succeed. Skeleton screens, not spinners.

The four-tier highlight hierarchy

Brand interactive color is 121 Group blue (#0077C8). It shows up at four intensities; pick the one that matches the weight of the action.

Tier 1 — Solid blue (CTA)

#0077C8 background, white text. Used for the single primary action on a surface: "Save changes," "Approve," "Request time off." One per screen. If two actions both want this tier, the design is unclear about which is primary.

Tier 2 — Blue ring (secondary)

1px #0077C8 border, transparent background, blue text. Used when a second action sits close to the primary in weight: "End early" next to "Approve," or "Skip" next to "Next."

Tier 3 — Blue tint (active state)

rgba(0,119,200,0.08) background in light mode, 0.14 in dark. Marks a current state: active tab, selected row, currently-on vacation banner, the "on" state of a toggle. Quiet but unmistakable when the eye finds it.

Tier 4 — Blue text (inline)

#0066CC (light) / #5AC8FA (dark). Inline links inside prose. Same treatment as a typical web link: change on hover, ring on focus, no underline by default.

Line length rule

Body text: 45-70 characters per line. Inter at 13-14px sits comfortably in this range when the column is around 480-720px wide. Headlines can run shorter (35-50 cpl) for impact. Short lines under 30 cpl read as fragments; long lines over 80 cpl break reading rhythm. When a card or panel is wider than 720px of body text, set a max-width on the prose.

When to break the rules

The principles are not laws. They're the default answer. Break any of them if you have a specific reason. Good reasons: the content type genuinely needs a different pattern (e.g. a creative tool needs more ornament, not less); user research shows the principle doesn't fit your audience; a regulatory or platform constraint forces a different choice. Bad reasons: "it looks cooler this way," "everyone else does it like this," "the stakeholder wants it," "we don't have time to do it right." If you break a rule for a bad reason, you're building friction into the product. Document the call so the next person knows it was a choice, not an accident.

Spacing tokens

Not strictly 4/8 grid. The values 7, 9, 10, 14, 18, 22, 26 are intentional — they smooth out vertical rhythm in dense UI where 8/16/24 jumps feel chunky. Stay on the named tokens; don't invent new ones. If you need a new spacing value, add a token, don't hardcode.

Token	Value	Use
--sp-hair	1px	Hairlines, divider widths
--sp-nano	2px	Pip spacing, 1/4 of base
--sp-micro	4px	Tight icon+label gaps
--sp-xs	6px	Compact chip/button padding
--sp-7	7px	Tight pill-button vertical padding
--sp-sm	8px	Standard tight gap (Apple 8pt base)
--sp-9	9px	Form input vertical padding
--sp-md	10px	Medium chip/button padding
--sp-base	12px	Section internal spacing
--sp-lg	14px	Card internal padding
--sp-xl	16px	Standard content padding
--sp-18	18px	Card/section internal
--sp-2xl	20px	Relaxed card padding
--sp-22	22px	Section horizontal
--sp-3xl	24px	Section outer padding
--sp-26	26px	Wide section horizontal
--sp-28	28px	Between-section gap (medium)
--sp-4xl	32px	Between-section gap
--sp-36	36px	Page-level outer padding
--sp-5xl	40px	Large section gap
--sp-6xl	48px	Hero padding / page-level rhythm
--sp-7xl	64px	Top-level view padding

Typefaces

Three font families show up in the build: the system stack for body, an SF Mono fallback for code, and Inter for specific overlay surfaces. The body is the default; everything else is opt-in.

Body — system font stack

Stack: `-apple-system`, `BlinkMacSystemFont`, `'SF Pro Display'`, `'SF Pro Text'`, `'Segoe UI'`, `sans-serif`. Set on the body selector; everything inherits unless it explicitly overrides. On macOS/iOS this resolves to SF Pro; on Windows to Segoe UI; on Android to Roboto via the system fallback. The deliberate choice: the app feels native on each OS, no flash of unstyled text on first load, no extra HTTP request before first paint. Most of the UI lives here.

Mono — SF Mono fallback chain

Stack: `'SF Mono'`, `Monaco`, `monospace` (the dominant pattern; a few surfaces use the slightly longer `ui-monospace`, `SFMono-Regular`, `Menlo` variant). Used for: code blocks in the card description editor, ID badges (e.g. `"#1142680"` on service breadcrumbs), kbd shortcuts (the `cmd-K` hint in search), and inline code in markdown notes. Same native-on-each-OS fallback strategy.

Inter — loaded but reserved for specific surfaces

Inter is loaded via Google Fonts in `index.html` at weights 400, 500, 600, 700. Currently used only in the testers session UI (the sign-in gate and identity widget that overlays the app before auth resolves). Available everywhere if you need it; the dev can opt-in for a specific surface by setting `font-family: 'Inter', -apple-system, ...` explicitly. Don't swap it onto body without a reason; the system stack is the default for a reason.

When to reach for what

Default: do nothing. The body inherits the system stack and you get OS-native typography for free. Code / IDs / shortcuts: use mono with the fallback chain above. Brand surfaces or specific moments where you want a consistent look across all OSes (e.g. an in-product onboarding screen): opt into Inter explicitly. The rule of thumb: one typeface for everything UI, two at most if there's a deliberate reason.

Type scale

Weight + size carry hierarchy before color or borders. The full type scale is below; the typeface section above explains which font the body, mono, and accent surfaces resolve to.

Sizes

Token	Value	Use
--fs-nano	9px	Avatar +N pips, AM/PM flag
--fs-micro	10px	Tiny badges, counts
--fs-xs	11px	Micro meta, kbd, small labels
--fs-sm	12px	Section labels, chip counts
--fs-md	13px	Body-small (most common; Inter optimal)
--fs-base	14px	Body default
--fs-lg	16px	Body-large, sidebar empty titles
--fs-xl	18px	H4, logo
--fs-2xl	21px	Section hero
--fs-3xl	24px	H2
--fs-4xl	28px	Hero name (client detail)
--fs-5xl	32px	Page title (overview, clients)
--fs-6xl	36px	Big stat

Line heights

Token	Value	Use
--lh-hug	1	Icons, numbers (no descender space)
--lh-heading	1.15	Page title
--lh-subheading	1.2	H2-H3, hero
--lh-dense	1.3	Compact body
--lh-body	1.4	Body default (labels, UI text)
--lh-relaxed	1.45	Long-form body, markdown

Color tokens

Brand interactive accent

Token	Value	Use
--highlight	#0077C8	121 Group blue. The brand interactive color.
--highlight-soft	rgba(0, 119, 200, 0.08)	Hover/active tint (light mode).
--highlight-soft (dark)	rgba(0, 119, 200, 0.14)	Slightly stronger tint for dark mode.
--link-blue	#0066CC	Inline links on light bg.
--link-blue (dark)	#5AC8FA	Inline links on dark bg (Apple Light Blue).

Surface

Token	Value	Use
--bg	#FFFFFF	Page background (light).
--bg (dark)	#000000	Page background (dark).
--bg-soft	var	Slightly elevated surfaces.
--bg-elev	var	Modal/popover surfaces.

Semantic

Token	Value	Use
Danger	#D33B3B / #FF6961 (dark)	Errors, destructive actions, blocked states.
Success	#4A8F4A / #30D158 (dark)	Confirmations, approved states.
Warning	#D69D2A / #FF9F0A (dark)	At-risk, due-soon, partial states.

Motion tokens

Always honor prefers-reduced-motion: if the user has it set, either skip the animation entirely or compress it to 1ms. The ease-out family is the default; pop and spring are reserved for moments where the motion teaches the user something (a popover appearing, a panel sliding from the side).

Durations

Token	Value	Use
--dur-instant	100ms	Color/bg flicks, immediate feedback
--dur-fast	120ms	Hover state changes (dominant pattern)
--dur-base	150ms	Standard UI transitions
--dur-med	180ms	Toast, menu, popover enter
--dur-slow	220ms	Larger transforms
--dur-slower	300ms	Panel slides, theme crossfade
--dur-page	400ms	Full-page transitions

Easing

Token	Value	Use
--ease-out	cubic-bezier(0.2, 0, 0, 1)	Apple standard ease-out (default)
--ease-snap	cubic-bezier(0.2, 0.8, 0.2, 1)	Snappy ease-out for confident moves
--ease-pop	cubic-bezier(0.2, 0.9, 0.3, 1.2)	Pop with overshoot (delightful entry)
--ease-spring	cubic-bezier(0.32, 0.72, 0, 1)	Apple spring (panel slides)

Component patterns

Concrete patterns we ship across the app. Each one is a small recipe; mix and match for the surfaces you build.

Buttons (3 tiers)

Primary: solid blue background, white text. One per surface. Used for the action the user came here to do.

Secondary: 1px blue ring, transparent background, blue text. Sits next to the primary when there's a peer action.

Tertiary: text-only, no border, hover tint on background. Quiet actions that don't need to compete.

Modals

Backdrop: `rgba(0,0,0,0.4)` with `backdrop-blur`. Fixed inset-0, z-index above topnav. Always honor Escape (close), click-outside (close), and the explicit close button. Focus trap inside; focus first input or close button on open. Lazy-load heavy modals via `React.lazy` + `Suspense`. Modal width caps: card modal around 720px, settings modal around 960px.

Dropdowns / popovers

Local state for open/closed. Click-outside listener attached when open, removed on close. Use `mousedown` not click on the toggle button so it doesn't fire-and-close on the same event. Use position: fixed and recompute top/left from the trigger's bounding rect on open, so horizontal scroll doesn't drift the menu. Dismiss on Escape.

Chips and pills

Two flavors: status chips (DONE/OPEN/PARTIAL with semantic colors) and count badges (e.g. '12' next to a tab). Always use named color tokens. Chips don't get drop shadows, they get a soft tinted background. Always pair color with a label or icon; never rely on color alone.

Card surfaces

Kanban tile: 14px padding, 8px radius, hairline border, subtle hover lift (`transform: translateY(-1px)` + shadow). KPI card: 20px padding, 12px radius, label + value + foot text, click expands an inline drill-down panel. List row: hairline divider between rows, 12-14px vertical padding, hover tint, click goes to detail.

Inputs and forms

Always pair an input with a visible label. Placeholders don't count as labels. Focus state: 2px blue outline with 2px offset. Disabled states: 0.5 opacity, cursor: not-allowed. Validation: inline error text under the field, never a popup.

Icons

Default to `@heroicons/react/24/outline`. Solid variants only for filled-state toggles (e.g. the kanban pin badge when active, the favorite star when starred). Common sizes: 14px for inline-with-text, 16px for buttons, 20px for nav rails, 24px for empty-state heroes. Never mix icon families on the same surface.

Empty states

Quiet, not punishing. One-line plain copy explaining what's missing and a single CTA showing how to fill it. Example: 'No boards favorited yet. Open any client and tap the star on a service to pin its board here. → Browse clients.' Don't use illustrations unless they pull weight.

House conventions

Specific calls we've made that aren't obvious from the principles. Match these to keep the build consistent with what's already shipped.

No pencil icons for editable fields. Cursor change + hover tint + focus ring carry the affordance. Pencil icons train users to expect a separate edit screen; click-to-edit-in-place is faster.

"+ Add card" only in the To Do column. Adding to In Progress or Done implies skipping the planning step. The compose flow lives only on the To Do column; if a card needs to land elsewhere, drag it after creation.

Click-to-edit for inline fields. Title, due date, assignee, labels — all click-to-edit. Modals only for actions that span many fields, multiple entities, or destructive confirmations.

Kebab menu for card-level actions. Duplicate, copy link, archive, delete — all live in the kebab menu. Promote out of the kebab only when the action is common enough to deserve a one-tap surface (e.g. the WIP pin we promoted to a header button).

Semantic color first, decorative color second. Red means danger. Green means success. Amber means warning. Don't use red for a non-destructive action just because it needs to stand out; use the tier-1 blue CTA instead.

Splice-replace on array mutations. {...original, ...patch} + concat for inserts. Never `Object.assign(arrItem, patch)`. Memoised consumers cache stale data otherwise.

No em dashes in copy. House style. Use commas, semicolons, or full stops. The em dash is the one punctuation mark we don't ship.

The 10-question audit

Run this before shipping any screen, modal, or feature. If the answer to any item is "no" or "not sure," fix it or document why you're skipping it on purpose. Severity tags flag what blocks shipping vs. what's polish.

- ☐ **HIGH** Purpose. A new user can tell what this screen is for in 5 seconds.
- ☐ **HIGH** Hierarchy. The most important thing on the screen is also the most visually prominent.
- ☐ **HIGH** Clarity. Every button, link, and field has one obvious meaning. No tooltips required to interpret a control.
- ☐ **HIGH** Feedback. Every action gives a visible response within 100ms. Anything over 200ms shows a loading state.
- ☐ **HIGH** Forgiveness. The user can undo anything they did by accident. Destructive actions confirm.
- ☐ **HIGH** Consistency. This screen uses the same patterns, words, and icons as the rest of the app.
- ☐ **HIGH** Accessibility. The screen works with only a keyboard. Works with a screen reader. Honors prefers-reduced-motion.
- ☐ **MED** Speed. First paint under 1.5s on a slow connection. No layout shifts after initial load.
- ☐ **MED** Respect. The screen feels like it's working for the user, not making the user work for it.
- ☐ **MED** Belief. The screen helps someone get a thing done and gets out of their way.

Reference files in the dev build

Read these to see how the patterns plug together in real code. The repo at github.com/nikkortrinidad-cpu/flizow is public; you can browse without signing in.

src/styles/flizow.css

Every token (--sp-*, --fs-*, --highlight, --dur-*, --ease-*) lives here. ~13000 LOC, but the token block sits at the top (roughly first 250 lines). Search for the block comments labeled by section.

CLAUDE.md (repo root)

Project reference. Tech stack, folder structure, naming conventions, the 4-tier highlight hierarchy. Read this once when you start; reference back for anything domain-specific.

docs/feature-specs/icon-manifest.md

Auto-generated. Every Heroicon used in the live build, grouped by source file, with a flat alphabetical index at the bottom. Grep this when you need to match an icon.

src/components/TopNav.tsx, FlizowAccountModal.tsx, FlizowCardModal.tsx

Three reference components that exercise most of the design system. Read these to see how the patterns plug together: nav structure, modal shell, dropdown, chips, kebab menu.

src/pages/AnalyticsPage.tsx

Reference for KPI cards, drill-down panels, filter pills. Most of the layout/grid patterns live here in one file.

src/components/BoardFilters.tsx

Reference for the search + filter chip bar shared by Board and Ops surfaces.